Response to Reviewers

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Snowmelt onset from Sentinel-1 backscatter across 41 Alpine catchments: a seven-year validation against in situ snow sensors

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Dear Dr Lindqvist,

Thank you for handling our manuscript. The reviewers' comments led us to extend the validation from five to seven melt seasons, to separate wet-snow detection from the onset date itself, and to report how the method behaves under forest cover.

The table below summarises the main changes. In the replies, each reviewer comment is in a tinted box, and quoted manuscript passages carry a coloured bar with their page and line numbers. A tracked-changes version is a supplementary file. We hope the manuscript is now suitable for the *Journal of Mountain Hydrology*.

Yours sincerely,
Elena Marchetti, on behalf of all authors
Institute for Snow and Water Research, Bolzano

Summary of changes

SECTION	CHANGE	RAISED IN
2.2 Data	Validation extended to seven seasons (2019–2025) and 63 stations	R1.2
3.1 Method	Onset defined as three consecutive wet-snow acquisitions, not one	R1.3, R2.1
3.3 Method	Threshold sensitivity from -1 to -4 dB added (new Fig. 5)	R2.2
4.2 Results	Forest-cover stratification of errors (new Table 3)	R1.4
5 Discussion	Limits for wind-crusted and refrozen snowpacks stated explicitly	R2.3
4.4 Results	Radar onset compared with optical snow-cover loss	R2.5
Figures	Colour-blind-safe onset scale in Fig. 3; scale bar in Fig. 1	R1.5
Data	Code and onset maps archived with a DOI	R2.4

Reviewer 1

Overall positive; asks for a longer record and a forest analysis.

R1.1 The manuscript is clearly written and addresses a relevant problem. My main concerns are the length of the validation period and the treatment of forested pixels, detailed below.

Response. We thank the reviewer for this positive assessment. Both concerns are addressed in the replies to R1.2 and R1.4.

R1.2 Five melt seasons seem short to claim robustness, especially since 2022 was an exceptionally dry year in the Alps. Could the authors extend the record, or at least show results without 2022?

Response. We agree. Sentinel-1 data for 2019 and 2025 are now processed, which extends the record to seven seasons and raises the number of station-years from 212 to 371. We also report all statistics with 2022 excluded; the median absolute error changes by less than half a day.

> Revised manuscript, p. 5, ll. 118–123
>
> We use all Sentinel-1 interferometric wide-swath acquisitions from 1 October 2018 to 30 June 2025, covering seven melt seasons and 63 automatic snow stations (371 station-years). Excluding the anomalously dry 2022 season changes the median absolute onset error from 4.1 to 3.8 days (Table S2).

R1.3 A single wet-snow acquisition can be triggered by a short warm spell in mid-winter. How do the authors avoid declaring melt onset too early in such cases?

Response. This was a real weakness of the first version. Onset is now defined as the first of three consecutive acquisitions classified as wet, which at the six-day revisit means at least twelve days of persistent wet snow. The mid-winter false starts the reviewer describes drop from 14 % to 3 % of station-years.

> Revised manuscript, p. 7, ll. 171–175
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> Melt onset is the date of the first of three consecutive acquisitions in which backscatter falls more than 3 dB below the dry-snow reference. Isolated wet acquisitions are treated as transient mid-winter melt and do not set the onset date.

R1.4 Roughly a third of the study area is forested, where C-band backscatter is dominated by the canopy. Errors should be stratified by forest cover.

Response. We now stratify all errors by Copernicus tree-cover density in three classes (Table 3). Below 30 % cover, performance is unchanged; above 60 %, the median error rises to 7.9 days, and we recommend masking these pixels.

> Revised manuscript, p. 10, ll. 252–256
>
> Errors grow with canopy density (Table 3): the median absolute error is 3.6 days below 30 % tree cover, 5.2 days between 30 and 60 %, and 7.9 days above 60 %. We therefore mask pixels with more than 60 % cover in the published onset maps.

R1.5 Figure 3 uses a rainbow colour scale for onset dates, which is hard to read and not accessible to colour-blind readers.

Response. Thank you; Figure 3 now uses the perceptually uniform *batlow* scale, with ticks every ten days and month names on the colour bar. The same scale is used in the supplementary onset maps.

R1.6 Minor points: (a) the units of backscatter are missing in Equation 2; (b) “SWE” is used before it is defined; (c) the station map lacks a scale bar.

Response. All three are corrected: (a) Equation 2 now states dB, (b) snow water equivalent (SWE) is defined at first use on p. 2, and (c) a 20 km scale bar has been added to Figure 1.

Reviewer 2

Asks for a sharper definition, a sensitivity analysis and stated limits.

R2.1 The text sometimes uses “wet-snow detection” and “melt onset” interchangeably. These are different quantities and should be kept apart throughout.

Response. We fully agree and have revised the terminology across the manuscript. Wet-snow detection now refers only to the per-acquisition classification, and melt onset only to the seasonal date derived from it (see also R1.3). A sentence in the introduction makes the distinction explicit.

> Revised manuscript, p. 2, ll. 41–44
>
> We distinguish two quantities: *wet-snow detection*, a classification of each acquisition, and *melt onset*, a single date per pixel and season derived from the detection series.

R2.2 The -3 dB threshold is taken from earlier work at other sites. How sensitive are the results to this choice?

Response. We added a sensitivity analysis over thresholds from -1 to -4 dB in 0.5 dB steps (new Fig. 5). Errors are lowest between -2.5 and -3.5 dB and vary by less than one day within that range, so we keep -3 dB.

> Revised manuscript, p. 8, ll. 198–201
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> Varying the threshold between -2.5 and -3.5 dB changes the median absolute error by at most 0.8 days (Fig. 5); thresholds above -2 dB produce frequent false onsets from surface roughness.

R2.3 The discussion should state where the method is not expected to work, for example on wind-crusted ridges or after refreezing.

Response. We added a paragraph on these limits. Refreezing resets the detection series but not the onset date, which is the intended behaviour; wind crusts raise the dry-snow reference and delay detection by a median of 2.4 days at the five ridge stations.

> Revised manuscript, p. 13, ll. 318–322
>
> The method assumes a stable dry-snow reference. On wind-exposed ridges, crust formation raises this reference and delays detection by a median of 2.4 days; users should interpret onset dates on such terrain as upper bounds.

R2.4 Will the processing code and the onset maps be made available?

Response. Yes. The code and the seven annual onset maps are archived on Zenodo under an open licence, and the Data Availability statement gives the DOI.

R2.5 The comparison with the optical snow-cover product in Section 4.4 is interesting but rather brief. Does the radar onset precede the loss of optical snow cover consistently?

Response. It does, by a median of 23 days across all stations, which is the expected interval between the onset of melt and the disappearance of the snowpack. We expanded Section 4.4 with this result and a short explanation.

> Revised manuscript, p. 11, ll. 281–285
>
> Radar melt onset preceded the date of optical snow-cover loss at 96 % of station-years, by a median of 23 days (interquartile range 15–31 days), consistent with the time a ripe snowpack needs to melt out.

We thank both reviewers once more; their comments have made the paper more careful and more useful.